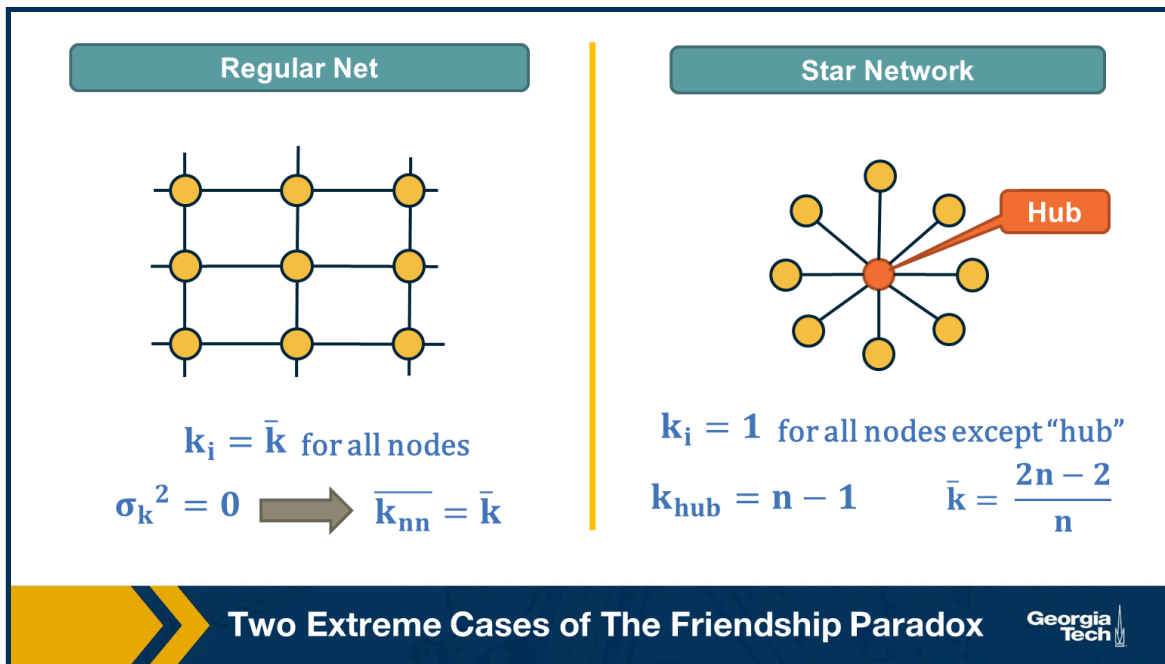


L3: Two Extreme Cases of The Friendship Paradox



Think of two extremes in terms of degree distribution: an infinitely large regular network in which all nodes have the same degree (*and thus the degree variance is 0*), and an infinitely large star network with one hub node at the center and all peripheral nodes connecting only to the hub.

In the regular network, the degree variance is zero and the average neighbor's degree is not different than the average node degree.

In the star network, on the other hand, the degree variance diverges as n increases, and so does the difference between the average node degree and the average neighbor degree.

Food for Thought

Derive the second moment of the degree distribution for the star network as the size of the network tends to infinity.